A 209-state candidate for the consistency of Peano arithmetic

Comparing three U211 proposals and combining
their state deletions through coordinated source reassociation

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29 October 2026

Abstract

The three current U211 proposals in ProveIt are distinct machines, including under state renaming. They implement the same ordered 3,909-primitive source program, but delete different states from their common U212 parent. Their patches cannot simply be combined: source reassociation changes the predecessor bits on which their local proofs depend.

We construct a complete binary, one-tape machine with **209 working states and 418 transitions**, including blank initialization. Thirty closed source reassociations expose compatible predecessor-bit invariants. Two existing selectors then support the simultaneous deletions of native states 180 and 181, and a transported continuation permits deletion of native 218. A ten-case whole-tape proof gives a positive block simulation with at most five child steps per step of the final rotated intermediate. Every primitive effect and resolved source control edge is preserved. Fresh symbolic verification proves all 5,084 arbitrary-natural primitive contracts, covers every literal transition, and agrees with an independent exact blank-start audit.

The new argument composes with the retained numerical-calculus and unary theory U calibration to give the ordinary mathematical, PA-formalizable endpoint $\mathrm{PA} \vdash \neg\,\mathrm{Halt}(U_{209}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})$. This packet supplies no new Lean, Rocq, MM0, or exported PA kernel certificate. We also prove a separate obstruction: deleting the native dominated propositional row makes the remaining calculus's consistency provable in PA, so that apparent simplification loses the desired calibration.

Executable identity. Working states $0, \ldots, 208$, start state 0, head coordinate 0, all-zero two-way infinite tape, alphabet $\{0, 1\}$, separate halt label 209, and only left/right moves.
Dense table SHA-256
`65ccb606c8e970adec22c311486e747b440171363601b8f11b56bc729cf8decb`

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1 Result and proof status

The requested threshold of fewer than 211 working states is met by two states. The primary executable artifact is `machine/u209.tm`. Appendix B prints all its transitions. The separate halting label is excluded from the working-state count; the complete initializer is included. Every counted state is reachable in the directed transition graph. This last assertion does not say that every state is visited on the blank run.

The proof uses inherited sparse *native labels*. In `machine/u209.native.tm`, 254 is halt and 211 is an ordinary working helper. Dense labels are obtained by sorting the 209 working native labels and assigning $0, \ldots, 208$; native 254 becomes dense 209. `machine/state_map.json` records both directions. Neither the large native label 254 nor the presence of native working label 211 changes the actual state count.

There are three logically different assertions in this report.

1. An unconditional finite-block simulation proves that the final rotated 212-state intermediate and U209 halt equivalently from state 0 on every common initial tape. Its proof needs only the literal transitions and their complete incoming-edge census.
2. The preserved source control-flow graph, final primitive runtime contracts, and identical complete startup configuration give blank-tape halting equivalence between U209 and the supplied U212 parent.
3. The supplied proof-search and interpretation arguments then identify that halting condition with a finite PA refutation.

Theorem 1.1 (Consistency endpoint, ordinary mathematical proof)**.** *For the exact dense table in this packet, composing the new machine/source proof with the retained U212 calibration gives*

$$\mathrm{PA} \vdash \neg\,\mathrm{Halt}(U_{209}, \mathrm{blank}) \iff \mathrm{Con}(U) \iff \mathrm{Con}(\mathrm{PA}).$$

The retained calibration is a substantive mathematical dependency, explained in Section 6 and included transitively in the unchanged Contextual reference archive. It is not a new assumption that PA is consistent. The finite-trace, source, syntactic-interpretation, and induction arguments are PA-formalizable. Executing their Python checks is distinct from checking a formal PA derivation. We supply no new end-to-end proof-assistant theorem for this table, assert no unconditional nonhalting, and make no global minimality or discovery-priority claim.

The K215 results in ProveIt and the separate MM0 development concern a different table and ZFC endpoint [5, 6]. They establish relevant infrastructure and context; they do not certify U209 by replacing the theory name in an existing theorem.

2 Comparison of the three supplied proposals

2.1 A pinned, reproducible comparison

The incoming directory was read at ProveIt commit

`da4ab9dd68ccfc47ac852118601eaf1fdae53795`.

It contains the unary-theory packet and the three archives in Table 1 [1, 2, 3]. Their exact bytes, source URLs, sizes, and SHA-256 identities are retained in `inputs/provenance.json`; the archives themselves are in `reference/`. The comparison reads the tables directly from those archives, checks their parent identities, and resolves every guarded source successor.

Short name	Exact incoming archive
Contextual	`U211_ConPA_Contextual_2026-10-28.zip`
Research	`U211_ConPA_Research_2026-10-28.zip`
Rotated	`U211_ConPA_Rotated_2026-10-28.zip`

Table 1: The current three U211 proposals, all retained unchanged.

Unlike the eight U212 packets compared in those earlier reports, the current three U211 tables are genuinely distinct. This is stronger than unequal hashes: no root-preserving state bijection preserves their two labeled transitions, writes, moves, and halt. A breadth-first traversal forces such a bijection from state 0 and finds an injectivity conflict for every pair. The distinguishing paths are in `evidence/input_comparison.json`. This is a statement about transition-graph isomorphism, not a claim that the machines compute different consistency conditions.

	Contextual	Research	Rotated	New U209
Working states	211	211	211	**209**
Rotations from common U212	6	9	11	30
Deleted native states	180	181	218	180,181,218
Changed retained transition rows	16	19	19	38
Changed source addresses	549	936	3,096	3,096
Maximum source depth	42	42	41	42
Maximum source zero run	6	6	6	6

Table 2: All changes are measured against the same U212 table/source.

All four sources have 3,909 primitive occurrences, 197 distinct source subtrees, and 5,083 guarded control arrows. Every ordered opcode and operand is identical, as is the complete ordinal control-flow graph. Thus the improvement is physical sharing of control states under tape invariants, without reducing the source's distinct-subtree count or changing its logical proof search.

2.2 What each proposal contributes

Contextual. Six rotations make native 163 a decoder for the ordered pair $(90, 211)$, reached after a zero, while both entries to 180 write one. State 180 decodes $(152, 211)$. The shared read-one branch and helper 211 permit $180 \mapsto 163$, with a preceding-bit lookup on read zero. Changing decoder 7 also affects an arithmetic entry, so 171/0 must be repaired. This is the shortest recipe of the three, but its proof has an additional two-case engine-entry obligation.

Research. Nine rotations expose two decoders with a shared read-one continuation. It deletes $181 \mapsto 220$ using helper 129. Its report carefully distinguishes the native numerical calculus from the positive dyadic stream proposed in the unary-theory manuscript. That distinction is essential to the consistency proof.

Rotated. Eleven rotations arrange that every entrance to 218 writes zero immediately to the left of the head. Existing state 157 then performs the same two branches in three steps, allowing $218 \mapsto 157$.

Its local proof is especially economical: two cases, one predecessor invariant, no new selector-row patch. It changes more source addresses, but shortens the maximum source depth by one.

For an isolated local lemma, Rotated is the simplest starting point. For retaining the full inherited logical lineage, Contextual is the most useful dependency packet. No proposal is uniformly superior in recipe length, proof organization, and source depth. Their historical symbolic inference counts measure verifier work, not the machines' actual running-time complexity, so we do not use those numbers as a performance ranking.

All three supplied quick reconstruction/audit commands were rerun successfully in separate working copies. We distinguish those fresh quick checks from the original full runtime receipts. The final U209 runtime was freshly regenerated over all its cases.

3 New construction: coordinating three incompatible savings

Let A be the frozen common U212, R the final rotated and repaired 212-state intermediate, and C the final U209. The construction has a source stage and a state-fold stage. `checks/build.py` replays a fixed recipe; it performs no search and uses no external solver.

3.1 Closed source rotations

The source is a binary tree. Its leaves are increment CI, guarded decrement CD, forward branch CB, backward branch CC, or halt CH, with the inherited operands. A decoder node normally reads one address bit without changing it and moves right. For closed nonhalting factors X, Y, Z, reassociation replaces

$$((X, Y), Z) \longleftrightarrow (X, (Y, Z)).$$

Here "closed" means that internal control cannot leave the factor except by ordinary sequential continuation. The source boundary analysis computes its forward escape depth, backward escape depth, and halt occurrence; all three factors have boundary $(0, 0, \text{false})$.

At each physical rotation the inner decoder has exactly one incoming edge, the selected edge from the outer decoder. Both rows of each are symbol-preserving right moves. The source checker locates every occurrence of that outer subtree, changes its address prefixes, and checks the ordered primitive word and all guarded ordinal successors. Two successor algorithms are used: subtree intervals and binary address increment/cut.

Preserving only leaf order would be insufficient: the branch operands refer to enclosing source positions. We verify their resolved targets after every rotation. The primitive counts are 1,558 increments, 1,175 decrements, 80 forward branches, 1,095 backward branches, and one halt. The decrements have separate zero and positive contracts, giving $3909 + 1175 = 5084$ runtime cases.

3.2 Why direct patch stacking fails

Start with Contextual's six rotations and Research's first eight. Research's ninth rotation is already represented in the Contextual construction and cannot simply be repeated. The resulting fourteen-rotation layout preserves source semantics but gives both candidate states 181 and 220 predecessor tag one. The old Research selector proof therefore has a false premise. The Contextual $180 \mapsto 163$ fold remains available; a second fold requires a changed association.

Four additional left rotations,

$$(217, 215, 0), \quad (215, 129, 0), \quad (129, 204, 0), \quad (204, 214, 0),$$

move 220 to a zero-writing entry and expose helper 215 with branches to 158 and 129. On the necessary read-zero return, those states lead to 218 and 147 respectively. This yields a valid 210-state intermediate with $180 \mapsto 163$ and $181 \mapsto 220$. Its full runtime was also checked during discovery, but it is not the principal delivered candidate.

3.3 Transporting a continuation to enable the third deletion

The remaining obstruction is the one-writing entrance $216/1 \to 218$. The rightmost 218 is a right child in the source layout. Appending an existing following factor to that region, by rotating it down the right spine, changes that entrance to a zero-writing one while retaining the zero tag of 163.

This continuation is the unchanged factor whose decoder label is 116. Twelve further closed rotations move it from the enclosing region down to the pair containing 218. They are steps 19–30 in Appendix A. The resulting relevant decoder pairs are

$$\begin{array}{lcllcl} 217 & = & (158, 215), & 215 & = & (147, 129), \\ 158 & = & (218, 178), & 220 & = & (218, 176), \\ 181 & = & (147, 176), & 163 & = & (90, 211), \\ 180 & = & (152, 211), & 216 & = & (218, 116). \end{array}$$

Each pair (s, t) denotes the two actions $0\mathrm{R}s$ and $1\mathrm{R}t$, before folding. The second selector must now be 217, not its former label 215. This adjustment is part of the new construction, not an unverified reuse of Research's original helper.

All four incoming edges to 218 now write zero. Thus the Rotated proposal's local $218 \mapsto 157$ mechanism is compatible with both other deletions. The final changes to R are precisely

$$\rho(180) = 163, \qquad \rho(181) = 220, \qquad \rho(218) = 157,$$

with every other state fixed; all transition targets are mapped through ρ, those three source states are deleted, and

$$\delta_C(163, 0) = (0, \mathrm{L}, 211), \qquad \delta_C(220, 0) = (0, \mathrm{L}, 217).$$

Before folding, the inherited arithmetic entry is repaired by $\delta_R(171, 0) = (0, \mathrm{L}, 205)$.

3.4 Intermediate addresses are not runtime claims

Seventeen of the thirty intermediate source layouts contain a run of seven zero address bits. Those intermediate assembled tables are not claimed to implement their source correctly under the seven-zero origin detector. They are construction stages for a finite source transformation. The final layout again has maximum zero run six and maximum depth 42.

The proof uses source graph preservation throughout, followed by the final literal runtime proof and exact startup. It does not compose thirty unproved claims of physical runtime equivalence. Independently, all 3,908 final internal source prefixes were executed with opaque exterior tape: their terminal states agree with the prefold states under ρ, and every inspected bit is preserved.

4 A complete local proof for the simultaneous fold

4.1 Entry invariants from a complete incoming census

Suppose an R-run begins in state 0 on any binary tape. Whenever its current state is one of the five special states in Table 3, the cell immediately left of the head contains the listed bit. This follows from the literal incoming transitions: every one writes that bit and moves right. None of the five states is initial. No assumption about the original contents of that cell is needed.

Old state	Predecessor bit	Complete incoming state/read pairs
163	0	$(214, 0)$
180	1	$(132, 1), (199, 1)$
181	1	$(185, 1)$
220	0	$(204, 0)$
218	0	$(158, 0), (211, 0), (216, 0), (220, 0)$

Table 3: Every listed transition writes the middle-column bit and moves right.

4.2 The ten exhaustive two-cell cases

Fix a head coordinate h, a current bit b, and the appropriate predecessor bit $t = T(h-1)$. The two machines agree on the full tape. Table 4 describes each old transition and its new block. All endpoints have head $h+1$, and the whole tape is unchanged. Only cells $h-1$ and h are read. All other cells, including an infinite exterior, are arbitrary.

Old state	New state	Tag	Read 0: endpoint/steps	Read 1: endpoint/steps
163	163	0	90 / 5	211 / 1
180	163	1	152 / 3	211 / 1
181	220	1	147 / 3	176 / 1
220	220	0	157 / 3	176 / 1
218	157	0	90 / 3	86 / 3

Table 4: Child endpoints are ρ of the old endpoints. Each old block has one step.

Two interactions deserve explicit attention. For old 163 reading zero, the child path, shown as state/head pairs, is

$$\begin{aligned}(163, h) &\to (211, h-1) \to (157, h) \\ &\to (67, h-1) \to (151, h) \to (90, h+1).\end{aligned}$$

It has five steps because the intermediate call to old 218 has itself been folded into 157. Treating it as the old three-step Contextual path would be incorrect.

For old 220 reading zero, the child path is

$$(220, h) \to (217, h-1) \to (158, h) \to (157, h+1).$$

The old one-step endpoint is 218, whose image is 157. The child block ends there; the following simulated old transition handles the behavior of 218. With predecessor bit one, the same selector

uses 215 and ends at 147, matching old 181. The remaining zero branch of the first selector uses $163 \to 211 \to 184 \to 152$. For old 218 reading one, the replacement path is $157 \to 148 \to 186 \to 86$, again with the appropriate head offsets. All writes equal the inspected bits, and none of these blocks visits halt.

Theorem 4.1 (Simultaneous fold on arbitrary initial tapes)**.** *The literal tables R and C halt equivalently from their common initial state 0 on every common initial binary tape.*

Proof. Relate (q, h, T) in R to $(\rho(q), h, T)$ in C, imposing the predecessor-bit condition only for the five special old states. It holds initially. Away from those states, the complete literal-table comparison verifies

$$\delta_C(\rho(q), b) = (w, d, \rho(z)) \quad \text{whenever} \quad \delta_R(q, b) = (w, d, z).$$

If the old successor is special, its incoming transition supplies the next predecessor bit. At a special state, Table 4 gives the corresponding positive block and the same tape/head at its endpoint. The old endpoint also satisfies the next invariant whenever required.

Induction on finite old runs yields aligned child block endpoints. If the old run halts, its matching child run halts. Conversely, if the child first halts after m steps and the old run had not halted within m steps, the simulation would construct at least m positive child steps with no earlier halt; an internal halt is excluded by the block certificates, and a halting endpoint would force the old endpoint to halt. This contradiction proves the converse. The argument uses finite induction, not an assumption about the existence of a completed infinite trace. For arithmetically encoded finite-support initial tapes, including blank, it is directly formalizable in PA. □

For a halting old run, write $N_{q,b}$ for its number of occurrences of state q reading b, and $N_{218} = N_{218,0} + N_{218,1}$. The exact cost identity is

$$T_C = T_R + 4N_{163,0} + 2N_{180,0} + 2N_{181,0} + 2N_{220,0} + 2N_{218}.$$

Hence $T_R \le T_C \le 5T_R$. These bounds concern *only* the final fold $R \to C$. Source addresses and their execution costs change earlier, so no such timing bound is asserted between the original A and C.

4.3 The repaired arithmetic entry

The reassignment of native 7 is inherited from Contextual and requires $171/0 : 0\text{L}7 \mapsto 0\text{L}205$. If the predecessor cell is zero, both old and repaired paths reach state 86 at the original head in two steps. If it is one, both reach state 164 one cell to the right in three steps. The zero just written by 171 is preserved. These are complete two-cell identities with arbitrary unread exterior, separately checked from the three state deletions.

5 Final source simulation and exact blank initialization

5.1 What the symbolic contracts quantify over

The final interpreter was checked for each primitive occurrence and each decrement guard. The input frame contains an arbitrary natural extra header gap, nine arbitrary natural logical register values,

and an arbitrary nonempty finite spare-register list. For a positive decrement the tested value is $1+n$; for its zero branch it is zero. The physical register permutation is

$$(6, 2, 0, 4, 3, 7, 8, 5, 1),$$

and the header-distance lower bound is 56. The source address is followed by zero padding, the header, and the unary-coded register file, with zero exterior on both ends. The invariant is exactly the inherited canonical frame, not an arbitrary tape assertion about source execution.

The checker represents run lengths and elapsed time by affine forms in thirteen slots: a constant, the extra gap, nine primary values, the first spare value, and the width of the remaining finite spare list. It uses three sound kinds of step:

1. A literal transition when the next represented run has positive constant lower bound.
2. Induction through an arbitrary-length same-state, same-direction run. This also handles a zero-length run without adding an unjustified step.
3. A finite-list traversal lemma, independently rederived from the literal table on an arbitrary unary block with opaque contexts.

The four list lemmas use the block words $B(v) = 0\,1^{v+1}$ and $C(v) = 1\,0\,1^{v}$. Their one-block proofs take $v+2$ steps and extend by list induction. The identity $C(L)1 = 1B(L)$, and its reversal, justifies the list normalization rule.

All run lengths must have nonnegative coefficients. No unknown possibly empty run is inspected by a literal step. Every successful contract has a positive affine elapsed time; the minimum constant is five. These are unbounded unary/list induction calculations, not 5,084 choices of small numeric register inputs. The two inference-engine files are byte-identical to the retained Contextual versions, and their hashes are checked on replay. They remain ordinary Python, within the stated mathematical trust boundary.

5.2 Fresh results and independent checks

Obligation	Result
Final working states / literal rows	209 / 418
Arbitrary-natural guarded primitive contracts	5,084 / 5,084
Literal rows covered by their derivations	418 / 418
Symbolic inference applications	1,250,049
Final source internal-prefix quotient matches	3,908 / 3,908
Universal simultaneous-fold branch cases	10
Universal arithmetic-entry repair cases	2
Deliberately damaged local controls rejected	13
Exact blank startup transitions	48,140
Nonzero cells at startup endpoint	120

The independent literal checker imports none of the builder, source utilities, symbolic runtime, optimizer, or saved verification results. Its checks use explicit exceptions and give identical results under normal and optimized Python. Its negative controls alter selector direction/write, the newly required selector helper, the 157 replacement, or incoming tag premises; each is rejected. The full runtime engine requires assertions enabled and refuses optimized Python.

Row coverage means that every transition participates in at least one checked symbolic derivation or list lemma. It is not a claim that every transition appears in the particular blank run.

5.3 The complete blank prefix

After exactly 48,140 transitions, both original U212 and U209 have native state 84, head -2, header cell 54, source ordinal 136, and address 001100110101010. In logical order $(N, T, W, A, D, H, X, Y, L)$, the store is

$$(1, 0, 13, 0, 0, 1, 0, 0, 1).$$

There are 43 spare registers of value one. The audit compares the complete finite set of 120 one cells, treating every other integer coordinate as zero. It also compares the whole literal transition trace, whose SHA-256 is

`8d9cef508b7a08195a64502866197cc05748e48e17142006512d8616f15481ef`.

No edited or deleted row occurs in this prefix. An independent source execution checks that this is the same continuation reached after 141 primitive steps from ordinal 106 and store $(1, 0, 4, 0, 0, 1, 0, 0, 1)$.

Proposition 5.1 (Blank reduction to the calibrated parent)**.** *The preserved ordinal graph, final primitive contracts, and exact shared startup establish, by finite source/run induction,*

$$\mathrm{PA} \vdash \mathrm{Halt}(C, \mathrm{blank}) \leftrightarrow \mathrm{Halt}(A, \mathrm{blank}).$$

Every nonhalting primitive returns to the expected canonical frame in positive finite time, and the unique halt primitive reaches the literal halt. The source graphs agree on ordinals and register effects. The exact startup aligns their initial continuation. These facts supply both directions; no finite simulation is used as evidence of actual nonhalting.

6 The inherited theory and proof-search calibration

6.1 Unary U and PA

The supplied theory [4] uses classical equality-free first-order logic over a nonempty domain, with membership as its only nonlogical symbol. For each formula $\varphi(x)$ whose only possible free variable is x, it has the schema

$$\begin{aligned}\forall a\, \exists b\, [\, &\forall x\, (x \in b \to \varphi(x)) \\ &\wedge \forall x\, (\mathrm{Sub}(x, a) \to (\varphi(x) \to x \in b))\,],\end{aligned}$$

where $\mathrm{Sub}(x, a)$ abbreviates $\forall z(z \in x \to z \in a)$. Bound variables are freshened, and the source a is not a free matrix parameter. "Unary" restricts free matrix roles, not the number of bound variables or the matrix's quantifier complexity. The cover may contain additional φ-objects; exact source-relative separation is not an axiom.

The supplied lower interpretation uses numerical representatives modulo coextensionality, with saturation ensuring the required membership congruence. Its dependency order matters. Local predecessor lemmas first give induction without assuming successor totality. The induction proof handles separately the case where no failure lies inside the given numerical object. Quantifying the

parameters of an induction assertion inside its unary matrix then gives full parameterized representative induction. Comparison and least numerical tuples provide bounded numerical comprehension. Only afterward are successors, numerical powers, saturated pair codes, and canonical finite histories obtained; iterating successor and addition interprets addition and multiplication.

In the other direction, PA interprets membership by binary-bit membership. Bitwise inclusion gives $x \leq a$, so induction through the finite range up to a constructs a cover for each fixed translated formula. This is formula-specific comprehension, not a uniform truth predicate. Freshening, substitution, relational term translation, and proof concatenation supply primitive-recursive finite-proof transformations. Their correctness is proved by syntactic induction in PA, yielding the asserted PA-provable consistency equivalence.

Our review found no concrete flaw in this argument, including its induction-before-successor bootstrap and its handling of numerical versus arbitrary graph parameters. The theory packet's six diagnostic suites were also rerun in normal and optimized Python. Those finite diagnostics do not replace the unbounded interpretation proof.

6.2 The numerical calculus actually executed

The literal machine executes the inherited native calculus $\mathcal{D}_U$, not the separate positive dyadic theorem stream described in the unary packet. Its numerical constructors use Cantor pairing P:

$$I(f,g) = 4P(f,g), \quad R(i,j) = 4P(i,j)+1, \quad Q(i,f) = 4P(i,f)+2, \quad E(i,j) = 4P(i,j)+3.$$

Truth is the alias 0 and the false target is 1. There are seven logical rows, one dominated propositional row, modus ponens, generalization, and the gated role-reassigned covering/vacuity bundle. The machine-facing cover permits matrix parameters and is calibrated through unrestricted weak covering W plus global irreflexivity. Its literal axiom set is different from unary U, while the finite-refutation translations preserve the required consistency strength.

The U222 calibration handles the small-code aliases, auxiliary equality through indistinguishability by membership, and the exact logical templates. The U214 emitter proof handles output/source collision $a = b$, codes 0 and 1, and the two freshness exclusions $c \neq a, b$. Completeness pads the matrix to a code $n > 1$, uses roles $(a, b, c) = (n, 0, 1)$, projects the cover, and restores variable names. Dropping those gates without a new proof would invalidate the atomic vacuity component.

6.3 The protected startup tail

Blank startup enters an interior axiom-builder continuation rather than a clean source root. The initial cache value 1 is an arithmetic parameter, not an already established theorem. The retained suffix proof produces nine genuine theorem heads above opaque numerical tail 4. That tail would be unsafe if exposed: $4 = P(1,1)$ and $1 = P(1,0)$.

The retained scheduler argument assigns conservative weights -1 to selection, 0 to modus ponens, and $+10$ to an ordinary completed axiom command. Starting at candidate 1, every command prefix has debt at most two. The nine initial theorem heads therefore retain a seven-head margin. A ghost explicit list length handles the alias $P(0,0) = 0$; numerical pairing alone is not used to infer stack length. Strong compilation of refutations works above arbitrary underlying stacks after a reset, successful candidate indices are cofinal, and every unsuccessful candidate calculation terminates. Thus the tail affects neither soundness nor completeness.

6.4 Where the full dependency resides

The Contextual reference archive retains this complete mathematical chain: its `baseline/selected_U212.zip` contains the selected U213 archive, which contains U214, then U218, then U222. The decisive source files are U222's `paper/report.tex` and retained `references/logic/` templates; U214's `proofs/role_emitter.tex`; and U213's `proofs/compiler.tex`, `proofs/source_transport.tex`, and `proofs/startup_suffix.tex`. Exact nested archive names are recorded in the packet README.

Consequently the dependency is not merely a citation to absent prior material. It is still inherited mathematics: this work has not rebuilt an end-to-end kernel proof of all those changed-theory layers. The complete composition underlying Theorem 1.1 is

$$\begin{aligned}\mathrm{Halt}(C, \mathrm{blank}) &\longleftrightarrow \text{calibrated source continuation halts}\\ &\longleftrightarrow \mathcal{D}_U \vdash 1\\ &\longleftrightarrow \neg\,\mathrm{Con}(W + \mathrm{IR})\\ &\longleftrightarrow \neg\,\mathrm{Con}(U) \longleftrightarrow \neg\,\mathrm{Con}(\mathrm{PA}).\end{aligned}$$

7 A new obstruction to deleting the propositional row

The word "dominated" does not mean logically redundant. It describes the numerical restriction $u \le a$ on one parameter of the propositional schema. Let $\mathcal{D}_U^-$ be the native calculus with that entire row deleted, retaining the seven other rows, truth, inference rules, and every enabled covering/vacuity bundle. Here $\mathrm{Con}(\mathcal{D}_U^-)$ means that code 1 has no derivation using exactly these retained primitive rules, not the ordinary classical consequence closure of the remaining formulas.

Theorem 7.1 (The row-deleted calculus is PA-provably consistent)**.** *PA proves that $\mathcal{D}_U^-$ has no derivation of target code 1.*

Proof. Define a Boolean valuation v on numerical formula codes by

$$\begin{aligned}&v(0) = 1, \qquad v(R(i,j)) = 0, \qquad v(E(i,j)) = 1,\\ &v(Q(i,f)) = v(f), \qquad v(I(f,g)) = [v(f) = v(g)].\end{aligned}$$

The last operation is Boolean equality, not material implication. This is a logical matrix for the reduced syntactic calculus, not an ordinary classical first-order model of U. The alias $I(0,0) = 0$ is compatible with the definition. Apart from that alias, recursive formula fields decrease below the positive constructor code. Bounded Boolean course-of-values recursion defines a primitive-recursive valuation, and PA proves its constructor equations by strong induction.

Truth is designated, modus ponens preserves value one, and generalization preserves the body's value. Native negation $N(f) = I(f,1)$ becomes Boolean negation, while $N(Q(i,N(f)))$ has the same value as f. Writing $e(x,y) = [x = y]$, the seven retained logical rows evaluate respectively to

$$\begin{aligned}&1, \quad e(e(B,C),e(B,C)), \quad e(1,e(0,0)), \quad e(C,C),\\ &\qquad e(1,e(0,0)), \quad e(C,C), \quad e(1,e(1,1)),\end{aligned}$$

all equal to one. Their exact native formulas are in the accompanying checker and detailed note.

For the role-a covering family, use its actual numerical definitions

$$\begin{aligned} e_0 &= R(c,a), & G &= Q(a, I(R(a,c), R(a,b))),\\ U_0 &= Q(c, I(e_0,a)), & L_0 &= Q(c, I(a, I(G,e_0))),\\ H &= Q(a, I(U_0, I(L_0,1))), & V &= I(R(a,b), Q(c, R(a,b))),\\ B^* &= N(I(V,H)). \end{aligned}$$

If $A = v(a)$, their values are

$$v(e_0) = 0, \quad v(G) = 1, \quad v(U_0) = v(L_0) = \neg A, \quad v(H) = 0, \quad v(V) = 1, \quad v(B^*) = 1.$$

This includes all numerical collisions and aliases. A disabled gate returns truth 0, also designated. PA verifies these fixed Boolean identities and inducts on a finite derivation to conclude that its last code has value one. Since $v(1) = 0$, target 1 is underivable. □

The deleted row has the formula

$$\Lambda(a,b,c,u) = I(I(I(c,u),a), I(I(a,c), I(b,c))).$$

Its value is $v(c)$ xor $v(u)$ xor $v(b)$, which need not be one. The permitted instance $a = b = c = u = 1$ satisfies $u \le a$, has native code 15,604,816, and evaluates to zero. It explicitly separates that row from all the retained rules and bundles.

If PA is consistent, a PA-provable equivalence $\mathrm{Con}(\mathcal{D}_U^-) \leftrightarrow \mathrm{Con(PA)}$ would contradict Gödel's second incompleteness theorem, because PA already proves the left side. Thus deleting the entire row cannot preserve the requested calibration without another substantive logical change. The finite cross-check evaluates 3,584 retained-row instances and 512 bundles; the unbounded theorem is the structural and derivation induction above.

This obstruction and the coordinated three-fold construction are new analysis relative to the three supplied packets. No broader historical novelty claim is made.

8 Reproduction, artifacts, and practical limits

The archive is self-contained for the new reduction. It needs Python 3.10 or later with the standard library; no network, solver, optimizer, proof-assistant installation, or downloaded executable is needed. From its top-level directory run

```
python3 -B reproduce.py
```

The default checks every manifest entry, compares the original archives, rebuilds both literal tables and source, runs the independent source/local/ startup audits, checks the row-deletion cross-check, and freshly regenerates all 5,084 symbolic runtime cases. It compares the complete regenerated case data with the retained records. Generated data goes to a temporary directory; use `-output /absolute/external/path` to retain receipts. `-quick` explicitly skips the unbounded runtime and reports that omission. It is not a replacement for the default check.

Compile the report in a working copy using

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

PDF metadata may vary across TeX versions. Verify the distributed manifest before editing or recompiling sealed files.

The key artifacts are the dense/native tables and state map in `machine/`; the thirty rotations and pinned parent in `inputs/`; independent and inherited checkers in `checks/`; freshly generated receipts and compressed per-case data in `evidence/`; and the unchanged four source archives in `reference/`. The latter preserve attribution and the complete selected mathematical dependency chain. Original licenses and notices are also retained under `licenses/`.

The construction achieves a state reduction; it is not a running-time improvement. The source still has 197 distinct subtrees, and the local folds introduce additional head movements. The search performed here does not establish a lower bound at 209. The concrete next formalization task is to port the ten local cases and final primitive certificates into the existing proof-assistant infrastructure, then connect the retained U/PA calibration to this exact table. That remains distinct work from the mathematical and executable verification delivered here.

References

[1] ProveIt incoming packet, *U211 ConPA Contextual*, 28 October 2026. Pinned GitHub archive. Retained unchanged under `reference/`.

[2] ProveIt incoming packet, *U211 ConPA Research: Decoder Reassociation*, 28 October 2026. Pinned GitHub archive.

[3] ProveIt incoming packet, *U211 ConPA Rotated*, 28 October 2026. Pinned GitHub archive.

[4] ProveIt incoming packet, *Unary power covering at PA strength*, 27 October 2026. Pinned GitHub archive.

[5] ProveIt, *Formal correctness of the exact 215-state machine*. Pinned verification directory.

[6] Elliot Glazer, *K215 halts iff ZFC is inconsistent (Metamath Zero, PA)*. `https://github.com/elliotglazer/k215-pa`. Read 29 October 2026; its checker was not rerun here.

[7] Elliot Glazer, *K230: a 230-state Turing machine for the consistency of ZFC*, including the Theory K description. `https://github.com/elliotglazer/k230-rocq`.

[8] Jeff's busy-beaver project, *A verified 254-state Turing machine that halts iff ZFC is inconsistent*; machine attributed there to Mira-Cult. `https://github.com/jeff87654/busy-beaver/tree/zf-254/ZF/m254`.

A Exact source rotation recipe

An entry (p, q, s) rotates outer decoder p around its inner child q on side s. Side 0 changes $((X, Y), Z)$ to $(X, (Y, Z))$; side 1 makes the reverse association. The full source addresses, factor sizes, and per-stage metrics are in `inputs/rotations.json` and `machine/construction.json`. The table also lists the three closed factor sizes, counted as primitive occurrences.

Step	Outer	Inner	Side	Closed factor sizes
1	216	220	1	56, 80, 27
2	204	216	0	136, 27, 159
3	216	163	1	27, 103, 56
4	206	199	1	81, 123, 26
5	205	206	1	80, 204, 26
6	7	205	1	29, 284, 26
7	215	204	1	160, 136, 186
8	217	181	1	52, 80, 160
9	185	217	0	132, 160, 482
10	174	185	1	162, 132, 642
11	129	158	0	26, 56, 78
12	217	215	1	160, 296, 186
13	204	214	0	52, 108, 136
14	215	129	0	26, 134, 296
15	217	215	0	26, 430, 186
16	215	129	0	134, 296, 186
17	129	204	0	52, 244, 186
18	204	214	0	108, 136, 186
19	162	203	0	503, 2,515, 78
20	203	223	0	401, 2,114, 78
21	223	188	0	56, 2,058, 78
22	188	177	0	1,042, 1,016, 78
23	177	89	0	80, 936, 78
24	89	174	0	294, 642, 78
25	174	217	0	26, 616, 78
26	217	215	0	134, 482, 78
27	215	129	0	52, 430, 78
28	129	204	0	108, 322, 78
29	204	214	0	136, 186, 78
30	214	216	0	130, 56, 78

B Complete dense transition table

In each action wDr, write w, move $D \in \{L, R\}$, and enter r. Entering 209 halts after the write and move. States 0–208 are working states. The initial state and head coordinate are 0; every tape cell is initially 0. This is exactly `machine/u209.tm`.

State	Read 0	Read 1	State	Read 0	Read 1
0	1 R 92	1 R 0	105	0 R 21	0 L 10
1	0 L 1	1 L 2	106	0 R 61	1 R 42
2	0 L 3	1 L 2	107	0 R 121	1 R 8
3	0 L 5	1 L 2	108	0 R 101	1 R 122
4	1 R 6	0 L 4	109	0 L 208	0 L 95
5	0 L 9	1 L 2	110	0 R 124	1 R 104
6	0 R 6	0 R 10	111	0 R 115	1 L 72
7	0 R 190	1 R 159	112	0 R 165	1 R 181
8	0 R 14	1 R 13	113	0 R 160	1 R 184
9	0 L 15	1 L 2	114	0 R 148	1 R 146
10	0 L 16	0 R 10	115	0 R 11	1 R 129
11	0 L 26	1 R 128	116	0 L 131	0 L 1
12	0 R 19	1 R 20	117	0 R 39	1 R 72
13	0 L 89	1 L 31	118	0 R 61	1 R 132
14	0 L 11	0 R 22	119	0 R 61	1 R 133
15	0 L 23	1 L 2	120	0 R 70	0 L 16
16	1 L 1	0 L 1	121	0 R 14	1 R 132
17	0 R 24	0 R 32	122	0 R 21	1 R 134
18	0 R 98	1 R 113	123	0 R 139	1 R 189
19	0 R 28	1 R 29	124	0 R 137	1 L 20
20	0 R 30	1 R 31	125	0 R 130	1 R 180
21	0 R 24	0 R 59	126	0 R 204	1 R 156
22	0 R 33	1 R 22	127	0 R 75	1 R 176
23	0 L 34	1 L 2	128	0 R 75	1 R 206
24	0 L 35	0 R 0	129	0 R 203	1 R 157
25	0 R 36	1 R 25	130	0 R 12	1 R 141
26	0 R 37	1 R 38	131	0 R 30	1 R 69
27	0 R 39	0 L 40	132	0 R 39	1 L 124
28	0 R 41	1 R 42	133	0 R 153	0 L 16
29	0 R 43	1 R 44	134	0 R 154	0 L 16
30	0 R 45	1 R 46	135	0 R 88	1 R 136
31	0 R 21	1 R 47	136	0 R 197	1 R 140
32	0 R 48	1 R 32	137	0 R 24	0 R 48
33	0 R 32	1 R 33	138	0 R 99	1 R 37
34	0 L 74	1 L 2	139	0 R 172	1 R 148
35	1 R 35	1 L 1	140	0 R 96	1 R 159
36	0 R 50	1 R 36	141	0 R 175	1 R 166
37	0 R 17	1 R 111	142	0 R 12	1 R 159
38	0 R 13	1 R 52	143	0 R 88	1 L 166
39	0 L 11	0 R 36	144	0 R 88	1 R 158
40	0 L 4	1 L 53	145	0 L 152	1 L 163
41	0 R 24	0 R 25	146	0 R 84	1 R 88
42	0 R 13	0 L 16	147	0 R 159	1 R 12
43	0 R 55	1 R 56	148	0 R 83	1 R 143
44	0 R 57	1 R 58	149	0 R 144	1 R 155
45	0 R 24	0 R 54	150	0 L 66	1 L 141
46	0 R 60	1 R 61	151	0 R 150	1 R 170
47	0 R 62	0 L 16	152	0 R 159	1 R 114
48	0 R 25	1 R 48	153	0 R 97	1 R 97
49	0 L 35	0 R 63	154	0 R 49	0 R 32
50	0 R 16	1 R 50	155	0 R 192	1 R 188
51	0 R 147	1 R 143	156	0 L 195	1 R 195
52	0 R 67	1 R 68	157	0 L 187	1 R 96

State	Read 0	Read 1	State	Read 0	Read 1
53	0 L 40	1 L 53	158	0 R 168	1 R 179
54	0 R 22	1 R 54	159	0 R 19	1 R 171
55	0 L 11	0 R 50	160	0 R 164	1 R 187
56	0 R 60	0 L 16	161	0 R 159	1 R 96
57	0 R 69	1 R 70	162	0 R 125	1 R 164
58	0 R 61	1 R 71	163	0 R 97	1 R 162
59	0 R 54	1 R 59	164	0 L 190	1 R 142
60	0 R 49	0 R 25	165	0 L 66	1 R 173
61	0 L 28	1 L 72	166	0 R 146	1 R 172
62	0 L 117	1 L 30	167	0 R 96	1 R 201
63	0 R 73	1 R 63	168	0 R 161	1 R 143
64	0 R 97	1 R 150	169	0 R 148	1 R 87
65	1 R 185	1 L 53	170	0 R 161	1 R 96
66	0 R 144	1 R 145	171	0 R 181	1 R 122
67	0 R 76	1 R 209	172	0 L 193	1 R 60
68	0 L 197	1 R 31	173	0 R 145	1 R 88
69	0 R 41	1 R 62	174	0 R 152	1 R 202
70	0 L 89	1 R 78	175	0 R 140	1 R 84
71	0 R 55	1 L 79	176	0 R 175	1 R 75
72	0 R 80	1 R 81	177	0 R 126	1 R 169
73	1 R 0	1 R 63	178	0 R 138	1 R 116
74	0 R 74	1 R 82	179	0 R 97	1 R 51
75	0 L 145	1 L 141	180	0 R 168	1 R 51
76	0 R 85	1 R 86	181	0 R 17	1 L 131
77	0 R 20	1 R 38	182	0 R 143	1 R 149
78	0 R 89	1 L 132	183	0 R 76	1 R 116
79	0 L 108	1 R 56	184	0 R 106	1 R 124
80	0 R 49	0 R 33	185	1 L 40	1 L 109
81	0 R 91	0 L 42	186	0 R 175	1 R 156
82	0 R 178	1 R 77	187	0 R 182	1 R 183
83	0 R 96	1 R 12	188	0 R 66	1 R 205
84	0 R 88	1 R 97	189	0 R 202	1 R 198
85	0 R 207	1 R 21	190	0 R 84	1 R 191
86	0 R 116	1 R 100	191	0 R 157	1 R 186
87	0 R 174	1 R 167	192	0 R 64	1 R 7
88	0 R 102	1 R 103	193	0 R 150	1 R 127
89	0 R 80	1 R 79	194	0 R 18	1 R 94
90	0 R 101	1 R 196	195	0 R 150	1 R 173
91	0 R 105	1 R 106	196	0 R 104	1 R 96
92	0 R 65	0 R 0	197	0 R 90	1 R 88
93	0 R 27	1 R 107	198	0 R 156	1 R 200
94	0 R 183	0 L 208	199	0 R 140	1 R 123
95	1 L 109	1 L 95	200	0 R 150	1 R 112
96	0 R 19	1 R 110	201	0 R 151	1 R 199
97	0 R 60	1 R 60	202	0 L 201	1 R 168
98	0 R 68	1 R 135	203	0 R 197	1 R 28
99	0 R 93	1 R 108	204	0 R 163	1 R 193
100	0 R 105	0 L 208	205	0 R 146	1 R 177
101	0 R 117	1 R 118	206	0 R 161	1 R 194
102	0 R 71	1 R 58	207	0 R 45	1 R 100
103	0 R 57	1 R 119	208	0 L 4	0 L 208
104	0 R 21	1 R 120			